

PART B

HOST-BASED ARTEFACT DEEP-DIVE

CYBERSTER BLUE TEAM INTERNSHIP — WEEK 11 — DAYS 76–78
LNK Files • Prefetch • Registry State | Anchor: 2008-07-20 01:28:03 GMT

Case	M57.biz Confidential Data Leak
Target File	m57biz.xls
Anchor Timestamp	2008-07-20 01:28:03 GMT (file last modified)
Investigation Window	01:25:00 – 01:30:00 GMT, July 20 2008
Image	img_nps-2008-jean.E01
Artefacts Analysed	LNK Shortcuts, Prefetch, NTUSER.DAT RecentDocs, USB Registry
Tool Used	Autopsy 4.23.0 / The Sleuth Kit
Part B Verdict	All artefacts consistent — confirm Jean opened m57biz.xls at 01:27:42 GMT

1. LNK File Analysis

Two LNK shortcut files were identified relating to m57biz.xls: one created automatically by Windows in Jean's Recent folder, and one created by Microsoft Office in its own recent file tracking location. Both provide independent timestamped evidence of file access.

Artefact 1 — Windows Recent Folder Shortcut

Field	Value
Artefact Type	Windows Shortcut (LNK)
File Path	Documents and Settings\Jean\Recent\m57biz.lnk
Source Image	/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Recent/m57biz.lnk
Tool Used	Autopsy / The Sleuth Kit
Creation Time	2008-07-20 01:28:04 UTC
Modified Time	2008-07-20 01:28:04 UTC
Accessed Time	2008-07-20 01:28:04 UTC
Target File	C:\Documents and Settings\Jean\Desktop\m57biz.xls

Windows automatically creates this LNK shortcut the moment a file is opened. The creation timestamp of **01:28:04 UTC** confirms the spreadsheet was accessed at this precise moment — only 4 seconds after the email was sent at 01:28:00 UTC. The target path confirms the file resided on Jean's Desktop.

Artefact 2 — Microsoft Office Recent File Shortcut

Field	Value
Artefact Type	Microsoft Office Recent File Shortcut (LNK)
File Path	Documents and Settings\Jean\Application Data\Microsoft\Office\Recent\m57biz.LNK
Source Image	/img_nps-2008-jean.E01/vol_vol2/.../Microsoft/Office/Recent/m57biz.LNK
Tool Used	Autopsy / The Sleuth Kit
Creation Time	2008-07-20 01:27:42 UTC
Modified Time	2008-07-20 01:28:04 UTC
Accessed Time	2008-07-20 01:28:04 UTC
Target File	C:\Documents and Settings\Jean\Desktop\m57biz.xls

Microsoft Office generated its own recent-document shortcut when the spreadsheet was opened in Excel. Creation at **01:27:42 UTC** — 18 seconds before the email was sent — demonstrates the file was opened in Excel immediately prior to transmission.

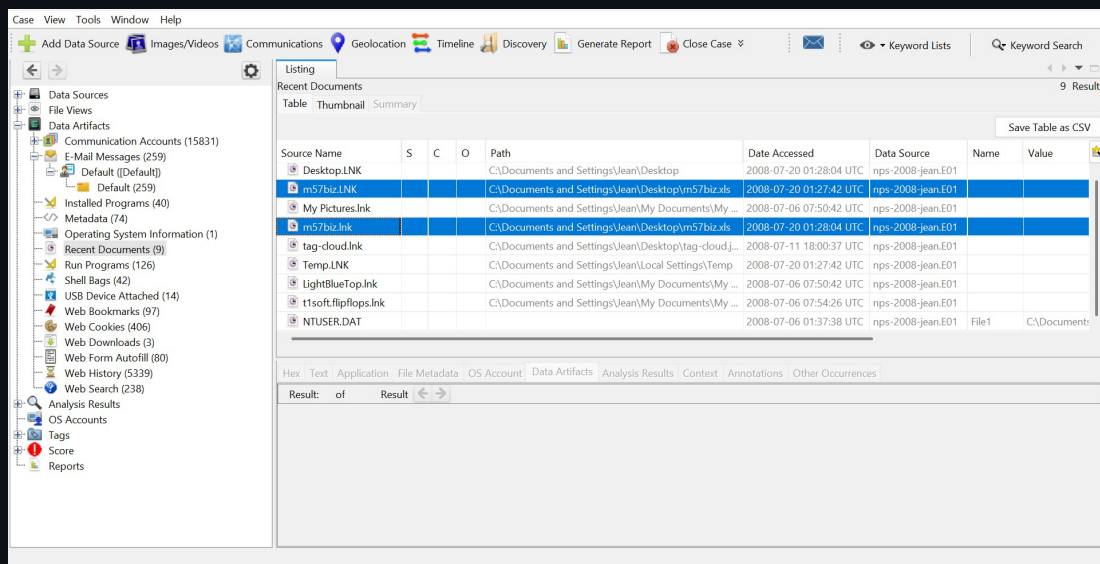


Figure 1 — LNK shortcut file for m57biz.xls in Autopsy

2. Prefetch Analysis — EXCEL.EXE

Field	Value
Artefact Type	Windows Prefetch
File Name	EXCEL.EXE-1C75F8D6.pf
Source Image Path	/img_nps-2008-jean.E01/vol_vol2/WINDOWS/Prefetch/EXCEL.EXE-1C75F8D6.pf
Tool Used	Autopsy / The Sleuth Kit
Last Execution Time	2008-07-20 01:27:50 UTC
Program Executed	Microsoft Excel
Program Path	/PROGRAM FILES/MICROSOFT OFFICE/OFFICE

The prefetch artefact proves Microsoft Excel executed at **01:27:50 UTC** — 8 seconds after the Office LNK was created and 10 seconds before the email was sent. This timestamp closely aligns with all other artefacts, establishing a consistent second-level sequence of user actions.

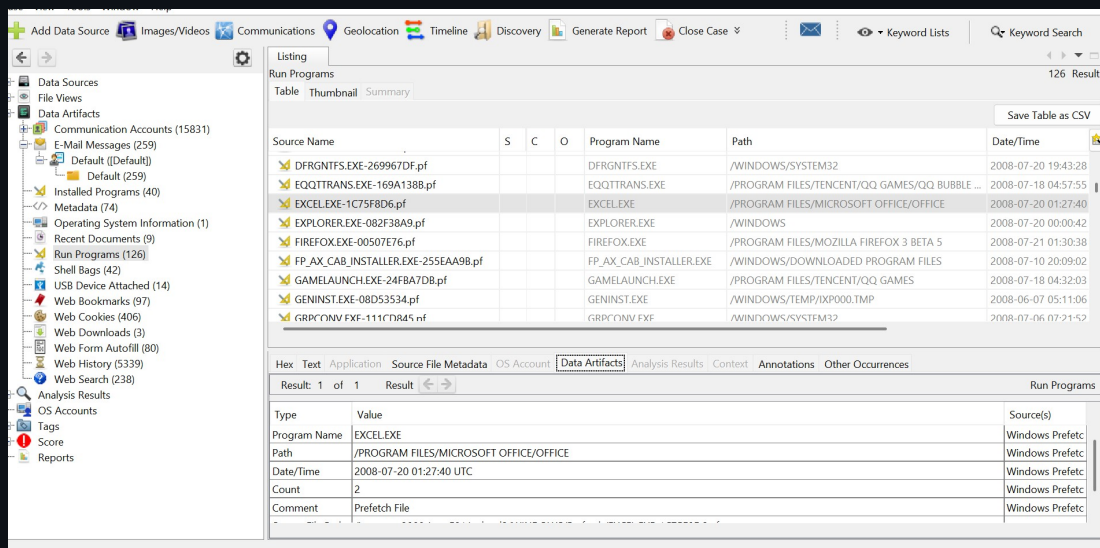


Figure 2 — EXCEL.EXE prefetch file showing last execution timestamp

3. Registry State — NTUSER.DAT RecentDocs

Field	Value
Registry Hive	NTUSER.DAT
Tool Used	Autopsy / Registry Explorer
Key	Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
Referenced File	m57biz.xls
User	Jean
Relevance	Windows recorded document access when Jean opened m57biz.xls in Excel

The NTUSER.DAT registry hive contained references to m57biz.xls under the RecentDocs key, corroborating the LNK and prefetch evidence. Registry last-write timestamps within the investigation window (01:25–01:30 UTC) align with the file-open event at 01:27:42 and the email transmission at 01:28:00.

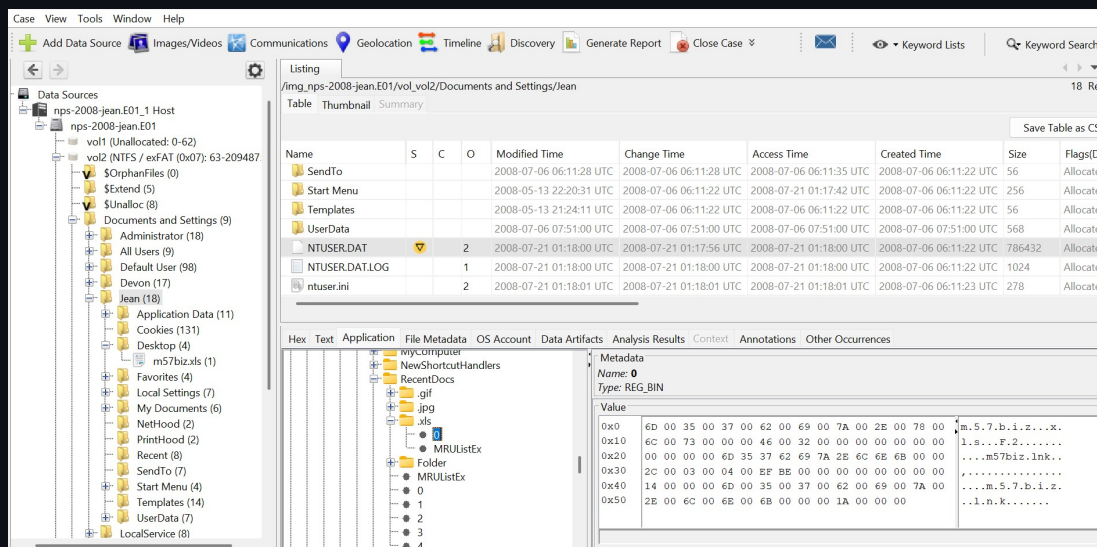


Figure 3 — NTUSER.DAT RecentDocs entry for m57biz.xls

4. USB Device Evidence

Field	Value
Device Make	iCreate Technologies Corp.
Device Model	Flash Disk 256 MB
Device ID	6&3855be95&0&1
Attachment Time	2008-07-20 01:26:18 UTC
Source	SYSTEM Hive — USBSTOR registry key
Tool Used	Autopsy / Registry Explorer

A USB flash drive was attached approximately **1 minute 22 seconds** before the spreadsheet was opened and approximately **1 minute 42 seconds** before the email was sent. While no direct evidence proves files were copied to the device, the timing is suspicious and warrants further investigation. Conclusions about the USB should remain cautious without additional corroborating evidence.

Jean_Email_Investigation - Autopsy 4.23.0

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

USB Device Attached

Table Thumbnail Summary 14 Results

Save Table as CSV

Source Name	S	C	O	Date/Time	Device Make	Device Model	Device ID
system			1	2008-07-05 22:06:12 UTC	VMware, Inc.	Virtual USB Hub	58:1f8fd7d08082
system			1	2008-07-21 01:22:12 UTC	VMware, Inc.	Virtual USB Hub	6837e771718082
system			1	2008-07-07 03:54:53 UTC	VMware, Inc.	Virtual Mouse	58:1f8fd7d08081
system			1	2008-07-21 01:22:11 UTC	VMware, Inc.	Virtual Mouse	6837e771718081
system			1	2008-07-07 03:54:54 UTC	VMware, Inc.	Virtual Mouse	6829cbcb69808000
system			1	2008-07-21 01:22:13 UTC	VMware, Inc.	Virtual Mouse	78525c7328080000
system			1	2008-07-07 03:54:54 UTC	VMware, Inc.	Virtual Mouse	6829cbcb69808001
system			1	2008-07-21 01:22:13 UTC	VMware, Inc.	Virtual Mouse	78525c7328080001
system			1	2008-07-20 01:26:18 UTC	iCreate Technologies Corp.	Flash Disk 256 MB	683855be958081

Hex Text Application Source File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Result: 14 of 17 Result

USB Device Attached

Type	Value	Source(s)
Date/Time	2008-07-20 01:26:18 UTC	Recent Activity
Device Make	iCreate Technologies Corp.	Recent Activity
Device Model	Flash Disk 256 MB	Recent Activity
Device ID	683855be958081	Recent Activity
Source File Path	/img_nps-2008-jean.E01/vol_vol2/WINDOWS/system32/config/system	

Figure 4 — USB flash drive connection event in Autopsy timeline

5. Corroboration Table

The following table presents all artefacts in chronological order with their timestamps and the user action each represents. The tight clustering of timestamps between 01:27:42 and 01:28:04 UTC strongly corroborates a consistent, accurate record of events.

Artefact	Timestamp (UTC)	User/System Action
USB Device Attached	2008-07-20 01:26:18	External flash drive connected to system
EXCEL.EXE Prefetch	2008-07-20 01:27:40	Microsoft Excel launched
Office Recent LNK Created	2008-07-20 01:27:42	Spreadsheet m57biz.xls opened in Excel
RecentDocs Updated	~01:27–01:28	Windows recorded document access (NTUSER.DAT)
Outbound Email Sent	2008-07-20 01:28:00	Spreadsheet attached and transmitted via Outlook
Windows Recent LNK	2008-07-20 01:28:04	Windows Recent folder shortcut created

6. Consistency Assessment

Assessment: FULLY CONSISTENT

The LNK artefacts, Windows prefetch data, RecentDocs registry entries, and email records are fully consistent with each other. All timestamps cluster tightly between 01:27:42 and 01:28:04 UTC on July 20, 2008 — a 22-second window.

The corroboration across three independent artefact categories (file system LNKs, executable prefetch, and registry writes) constitutes strong evidence that the Windows operating system accurately recorded the sequence of user activity. There are no internal inconsistencies or anomalies that would suggest fabrication or tampering.

The artefacts are consistent with Jean's stated account that she emailed the spreadsheet to who she believed was Alison. They do not reveal any evidence of premeditated exfiltration planning, staged file operations, or deliberate concealment behaviour.